Reproducible Synthesis of Long-Range Dependent Symbolic Sequences in Julia

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***Abstract*—Synthetic long-range dependent (LRD) data are useful only when the object being synthesized matches the object being tested. This is already difficult for numerical time series; symbolic data add another source of ambiguity. A DNA base, a word, or a security-event class has no natural numerical distance, yet many diagnostics and models quietly introduce one. This paper presents `SymbolicLongMemorySequences.jl` (Self-Similar Symbols Sequence Synthesis),[1] a Julia package for reproducible synthesis of LRD finite-alphabet sequences. The package combines property-based methods, which generate numerical LRD processes and then symbolize them, with model-based methods, which generate symbolic dependence directly through finite-state, renewal, excitation, or copy/mutation mechanisms. The paper takes an operational view: specify the alphabet, marginal frequencies, local structure when available, nominal LRD mechanism, random seed, and validation evidence. We give the mathematical construction of each method, report one-hot ACF and spectral diagnostics, test marginal control, and benchmark generation cost. The result is not a universal symbolic-LRD generator, but a reproducible workbench for estimator testing, controlled machine-learning stress tests, privacy-preserving simulation, and studies of symbolic data such as text, DNA, cyber-event logs, and network measurements.**

***Index Terms*—Long-range dependence, symbolic sequences, synthetic data, self-similarity, Julia, reproducible research.**

I. INTRODUCTION

Reproducible research needs data that can be shared, regenerated, and varied under controlled conditions. In many domains the real data are proprietary, privacy-sensitive, expensive to label, incomplete, or simply unavailable at the scale needed for controlled experiments. Synthetic data can provide ground truth, sensitivity checks, and repeatable comparisons [1], but only when the generator is explicit about what it controls and what assumptions it inserts. Network topology and traffic-matrix synthesis illustrate the same tension: a useful generator must be controllable enough for experiments, realistic enough not to mislead, and reproducible enough for other researchers to interrogate [2], [3].

Long-range dependence (LRD), or long memory, makes this need sharper. LRD means that observations far apart in time remain statistically coupled. In the covariance setting, a stationary process is often described as LRD when

$$\gamma(k) \sim c_\gamma k^{-\beta}, \qquad 0 < \beta < 1, \tag{1}$$

so the covariance tail is not summable. Equivalently, many familiar models have a low-frequency spectral pole, with spectral density proportional to $f^{-(1-\beta)}$ [4], [5]. Averages, counts, spectra, and estimators can therefore converge much more slowly than they would for short-memory data. Validation often needs long sequences and multiple replicates, which is exactly the regime where real traces are hardest to obtain and share.

The usual LRD story is numerical: river levels, traffic volumes, queue lengths, or other scalar time series. The focus here is different. Many data streams with plausible long memory are symbolic rather than naturally numerical: text, DNA, categorical event logs, cyber-security disclosures, packet-loss states, social-media events, and finite-state workflow traces. Generating such sequences supports several tasks:

- controlled finite-alphabet examples for LRD estimator development and finite-sample sensitivity studies;
- test sets for modeling estimators, including entropy and entropy-rate estimators, mutual-information diagnostics, and string-matching algorithms;
- stress tests for sequence models on non-language data with language-like long context; and
- privacy-preserving synthetic event streams when real traces cannot be shared.

Modeling the generation mechanisms can also sharpen scientific questions: which features of a sequence are explained by marginals, by local transition rules, by renewal structure, by latent numerical dependence, or by genuine long context?

Symbolic LRD is not obtained by simply attaching a Hurst parameter to an alphabet. A common shortcut is to map symbols to numbers and then use numerical LRD tools. That can be useful, but it introduces a metric that the alphabet may not possess. Mapping $\{\mathtt{A}, \mathtt{C}, \mathtt{G}, \mathtt{T}\}$ to $\{1, 2, 3, 4\}$, for instance, makes the numerical distance from `A` to `T` three times the distance from `A` to `C`. That geometry is an artifact of the encoding, not of the biology. Conversely, reducing a symbolic sequence to a few count series can hide dependence that lives in order, recurrence, or regime changes.

The task is therefore operational. A useful symbolic-LRD generator must answer more than "what is the nominal H?" It must specify the alphabet, the marginal distribution, any short-range structure that should be controlled, the mechanism that creates long memory, the finite memory or truncation used in

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[1]LRD and statistical self-similarity are closely related but not identical: LRD describes persistent dependence, often through a non-summable covariance tail or low-frequency spectral pole, while statistical self-similarity describes distributional invariance under aggregation or rescaling. Many classical models connect them through the Hurst parameter, but finite symbolic sequences require method-specific diagnostics.

practice, the random seed and provenance, and the diagnostic used to decide whether the generated sequence behaves as intended. A method may match a marginal distribution exactly while producing weak symbolic ACF diagnostics; another may have a convincing latent numerical LRD process that is attenuated by quantization; another may generate heavy-tailed recurrence times without a clean one-hot spectral pole. These are not minor implementation details. They are part of the scientific meaning of the synthetic data.

We present `SymbolicLongMemorySequences.jl`, a Julia package that makes these choices explicit. The package brings together property-based methods, which generate numerical LRD processes and transform them into symbols, and model-based methods, which build symbolic dependence directly through finite-memory, regime, renewal, self-exciting, or copy/mutation mechanisms. The central goal is reproducible experimentation: explicit constructors, explicit random-number generators, saved provenance, fast contract tests, opt-in validation campaigns, and benchmark artifacts. The package is not a universal symbolic-LRD generator. It is a workbench in which mechanisms can be constructed, compared, diagnosed, and rejected when they do not support the claim being made.

The contributions of this work are:

- a reproducible generator framework for finite-alphabet symbolic LRD sequences;
- a taxonomy separating property-based latent-process transformations from model-based symbolic mechanisms;
- new and adapted symbolic synthesis methods, including finite-memory uses of LAMP-style processes, a scalable dyadic-history LAMP approximation, and self-exciting symbolic-mass variants motivated by the failure of a naive normalized Hawkes translation;
- explicit control contracts for alphabets, marginal distributions, local Markov structure, nominal LRD parameters, and finite memory limits;
- a complexity analysis linking generation cost to sequence length, alphabet size, history depth, and calibration iterations;
- validation scripts that diagnose both symbolic one-hot processes and, for property-based generators, the original numerical latent process.

Not every proposed model performs well, and this is part of the paper's argument. Negative and partially successful constructions are retained because they show where intuition from numerical LRD breaks down for symbolic data. The paper therefore treats validation evidence as part of the method, not as an afterthought.

II. Related Work

The LRD literature is mature, but not monolithic. Covariance definitions focus on the tail of $\gamma(k)$; spectral definitions focus on the low-frequency pole; aggregation definitions connect to statistical self-similarity; and point-process definitions often use count variance or recurrence-time tails [4]–[6]. The distinctions are not academic bookkeeping. A finite-state chain can have strong local persistence without asymptotic LRD, while a renewal mechanism can have heavy-tailed counts without giving a simple symbol-level covariance target. This is why the package is organized by mechanism rather than by a single advertised exponent.

Numerical LRD synthesis provides the technical starting point. Spectral fGn approximations [7], exact and approximate Gaussian simulation [8]–[11], wavelet and multiscale methods [12]–[14], intermittent maps [15], mixtures of short-memory components [16], and heavy-tailed on/off or fractal point-process models [17]–[19] all offer ways to make long memory. They also show the usual engineering tradeoff. Exact Gaussian methods give strong second-order guarantees but can be expensive or fragile for some covariance forms. Spectral methods are fast and transparent but approximate. Wavelet, Haar, and fast fBm estimators show how much effort has gone into measuring fractal parameters from noisy finite records [12], [13], [20]; generalized fBm/fGn constructions also show that even the numerical vocabulary has variants [21]. Renewal and on/off models are interpretable, but finite runs can be dominated by missing scales or realization-to-realization variability [22]. A symbolic package should expose these tradeoffs rather than hide them behind a constructor named after H.

Symbolic LRD appears most clearly in DNA and text. Voss's DNA analysis [23] is especially important for this paper because it uses base-specific indicator processes instead of imposing an arbitrary scalar encoding. Related DNA work studies nucleotide correlations, spectra, and copy/mutation mechanisms [24]–[27]. Text studies report long correlations, hierarchy, bursty word use, and rare-word clustering [28]–[34], with more recent work connecting long context to neural sequence models [35]–[37]. This literature is a useful warning: DNA walks, word counts, rare-word arrival processes, semantic embeddings, and one-hot indicators can all see different aspects of the same sequence.

Several symbolic models are close to the constructions used here. Linear additive Markov processes [38] and additive Markov-chain memory-function models [39]–[41] motivate finite-alphabet mechanisms in which weighted history affects the next-symbol distribution. Hawkes-process models for word occurrence [42] provide a bursty self-exciting route, though the categorical one-symbol-per-time setting changes the normalization problem. Duplication and expansion-modification models for DNA-like sequences [26], [43], [44] provide a growth mechanism. `SymbolicLongMemorySequences.jl` implements these ideas as transparent synthesis methods rather than fitted models: parameters are construction controls, not claims of inference from a real corpus. Some implementations are therefore new operational variants of older ideas, especially the dyadic-history LAMP approximation and the MB4b/MB4c self-exciting mass processes.

Finally, symbolic long memory also touches information-theoretic work on texts and Hilberg-like laws [45]. That connection is not the main focus of this paper, but it reinforces the same point: for symbolic data, the object of interest is often not one scalar time series but a collection of recurrences, counts, transitions, and transformations. A useful synthesis tool must make those choices visible.

III. Operational Design

The introduction frames symbolic synthesis as an operational specification problem. This section turns that frame into the package design: what must a user define before a synthetic symbolic sequence is useful for an experiment? Length and alphabet are not enough. The user must decide the observable symbol space, the intended long-memory mechanism, the marginal frequencies, any short-range structure that should survive generation, and the random source needed for reproducibility. We write a generator schematically as

$$g = (\mathcal{A}, n, \theta_{\mathrm{LRD}}, \boldsymbol{p}, \mathcal{L}, d, \texttt{rng}, \mathrm{method}), \tag{2}$$

where $\mathcal{A}$ is the ordered alphabet, n is the sequence length, θ_{LRD} is a method-specific long-memory parameter, $\boldsymbol{p}$ is an optional marginal distribution, $\mathcal{L}$ is optional local structure such as Markov transition matrices, d is a configured or effective memory cutoff when applicable, and `rng` records the reproducible random source. Not every method supports every control. The API therefore records supported controls rather than presenting all methods as interchangeable. Throughout the paper, bold symbols in math mode denote vectors; scalar components of those vectors are written with subscripts.

A. LRD Parameters and Self-Similarity

For a second-order stationary numerical process, the common covariance parameterization is

$$\gamma(h) \sim c_\gamma h^{-\beta}, \qquad 0 < \beta < 1, \tag{3}$$

where $\gamma(h) = \mathrm{Cov}(X_t, X_{t+h})$.[2] The corresponding low-frequency spectral form is

$$f(\lambda) \sim c_f |\lambda|^{-\alpha_s}, \qquad \alpha_s = 1 - \beta, \tag{4}$$

and the usual Hurst-parameter relationship is

$$H = 1 - \frac{\beta}{2} = \frac{1 + \alpha_s}{2}, \qquad 1/2 < H < 1. \tag{5}$$

Equivalently, $\beta = 2 - 2H$ and $\alpha_s = 2H - 1$. These equations are a translation guide among common numerical conventions. They are not a guarantee that a finite symbolic generator with a nominal H will display the same slope in every diagnostic.

The notation needs care because the literature reuses letters. In this paper α_s denotes a spectral exponent, while some generator descriptions use α for a heavy-tail exponent, for example the Pareto sojourn exponent in MB2. For ideal heavy-tailed on/off constructions with $1 < \alpha < 2$, the aggregate count-process heuristic is $H = (3 - \alpha)/2$, which is not the same relationship as $\alpha_s = 2H - 1$. The package documentation therefore explains parameters method by method rather than treating H, α, and β as interchangeable labels.

LRD is closely connected to statistical self-similarity. If a process is aggregated into blocks of size m, self-similar scaling means that suitably normalized block averages retain the same large-scale distributional form. For symbolic sequences the observable object may be a symbol-indicator process, symbol counts in blocks, recurrence times, or a latent numerical driver. For this reason `SymbolicLongMemorySequences.jl` reports mechanism, transformation, and diagnostic together: a symbol sequence can be self-similar in one operational view and almost invisible in another.

B. Marginal and Local Controls

The marginal distribution $p_j = \Pr(X_t = a_j)$ controls how often each symbol appears. It matters before we even discuss LRD. First, it determines the variance of one-hot diagnostics: rare symbols give noisy autocorrelations. Second, it is often a domain constraint. DNA is the clean example. Chargaff's rules concern the relative frequencies of A, C, G, and T, and real genomes have species-specific base compositions as well as differences between species [46]. A symbolic DNA generator that cannot control at least the marginal distribution is missing a basic biological feature. Text has an equally familiar version of the same issue: word frequencies often follow a Zipf-like distribution, so a synthetic text-like sequence with uniform word frequencies would miss a first-order property of the domain [47]. Similar concerns appear as event-frequency imbalance in logs. Third, uncontrolled marginals can make a synthetic benchmark too easy: a model may learn the symbol imbalance rather than the long memory.

Short-range structure is a different knob. A first-order Markov specification uses a transition matrix

$$P_{ij} = \Pr(X_t = a_j \mid X_{t-1} = a_i) \tag{6}$$

to prescribe local repetition, alternation, or symbol-to-symbol preference. In `SymbolicLongMemorySequences.jl` this appears directly in methods such as PB3 and MB2, and indirectly through transition or excitation matrices in additive and self-exciting models. Controlling P is not the same as controlling LRD: a finite-state Markov chain with a fixed transition matrix is short-memory under ordinary conditions. The long-memory mechanism must enter through a persistent latent driver, a heavy-tailed regime duration, a power-law history kernel, or another scale-free construction.

The design goal follows directly from the paper's narrative: simultaneous control with explicit limits. A useful symbolic LRD generator should let the user say: use this alphabet; make these symbols common or rare; preserve this local Markov behavior when the method supports it; and inject long memory through this documented mechanism. These controls can conflict. Strong marginal calibration can attenuate latent dependence; Markov regimes with identical stationary distributions can hide LRD from one-hot diagnostics; finite memory d can truncate a nominal power law. The package therefore reports both capability and caveat in constructors, documentation, validation plots, and provenance metadata.

C. Interface and Evidence

The public contract is intentionally small:

$$\texttt{generate(g, n; rng)} \rightarrow \boldsymbol{X}_{1:n}. \tag{7}$$

[2] We use $\sim$ in its standard asymptotic-equivalence sense: for positive functions f and g, $f(k) \sim g(k)$ as $k \to \infty$ means $\lim_{k\to\infty} f(k)/g(k) = 1$. Thus a relation such as $\gamma(k) \sim c_\gamma k^{-\beta}$ is a large-lag tail statement including its multiplicative constant, not a claim that finite validation plots exactly match a power law at every displayed lag.

Generators preserve alphabet element types and reject invalid probabilities, transition matrices, and parameter ranges at construction time. Convenience functions construct standard cases by method identifier, while method-specific constructors remain available when the scientific details matter. The short API supports routine experiments; explicit constructors keep the method assumptions visible.

Property-based generators also implement

$$\texttt{generate\_with\_latent}(g, n; \texttt{rng}), \tag{8}$$

which returns both the emitted symbols and the latent numerical matrix. The latent output is needed for validation. If the symbolic diagnostics look weak, it helps distinguish a weak numerical source from dependence lost in the transformation or from finite-sample variability.

The reproducibility story also requires separating correctness evidence from scientific evidence. Unit tests cover construction, reproducibility, type preservation, and finite-sample contracts. Validation scripts produce longer Monte Carlo diagnostics: one-hot autocorrelation, one-hot spectra, marginal control, local transition control, and comparison conventions compatible with LongMemory.jl. Benchmarks live in a separate project environment and run only on explicit request. This separation keeps ordinary package tests short while retaining the longer validation evidence used in this paper.

IV. Methods

The methods are grouped by how they create dependence. Property-based methods start with a numerical LRD process and then ask how much dependence survives the symbolizer. Model-based methods generate symbols directly, so their claims are tied to the symbolic mechanism itself. Table I summarizes the implemented methods, their main controls, computational cost, and the caveat a user should keep in mind before trusting a plot.

A. Property-Based Methods

Property-based methods are attractive because numerical LRD synthesis is well understood. They are also dangerous in a useful way: the transformation can change the dependence we thought we had. The package therefore treats the latent process and the symbolizer as separate objects.

Let $\mathcal{A} = \{a_1, \dots, a_K\}$ be the ordered alphabet and let $\boldsymbol{p} = (p_1, \dots, p_K)$ be a target marginal distribution where applicable. Property-based generators first create a real-valued latent matrix $Z \in \mathbb{R}^{W \times n}$ and then apply a symbolizer $T : \mathbb{R}^{W \times n} \to \mathcal{A}^n$. The implementation exposes this split through

$$(\boldsymbol{X}_{1:n}, Z) = \texttt{generate\_with\_latent}(g, n; \texttt{rng}),$$

so validation can ask whether the LRD signal is present in Z, lost in T, or only weakly visible after finite-sample averaging. This finite-record view is consistent with the signal-processing literature on estimating fractal and fBm parameters from noisy observations [12], [13], [20].

a) PB1: spectral fGn with quantization.: PB1 generates a single latent series Z_t using Paxson's approximate spectral fractional Gaussian noise method [7]. On the Fourier grid, the target spectral shape is

$$S(f_j) \propto |f_j|^{1-2H}, \qquad H \in (1/2, 1), \tag{9}$$

with random phases and Hermitian symmetry giving a real inverse FFT. The implementation normalizes the latent series to zero sample mean and unit sample variance. The symbolizer is deterministic rank binning: choose integer bin counts c_i with $\sum_i c_i = n$ and $c_i \approx np_i$, sort the latent values, and assign the smallest c_1 values to a_1, the next c_2 values to a_2, and so on. Thus PB1 has strong finite-sample marginal control, but short-range symbol structure is induced rather than directly prescribed.

b) PB2: latent Gaussian categorical model.: PB2 uses K independent fGn streams $Z_{i,t}$ with common Hurst parameter H. The emitted symbol is

$$X_t = a_{\arg\max_i \{Z_{i,t} + \theta_i\}}, \tag{10}$$

where offsets θ_i are entries of the offset vector $\boldsymbol{\theta}$, which is calibrated on the realized latent sample to approximate the target marginal distribution. `SymbolicLongMemorySequences.jl` uses the iterative update

$$\theta_i \leftarrow \theta_i + \eta \left[\log(p_i + \epsilon) - \log(\hat{p}_i + \epsilon)\right], \tag{11}$$

followed by centering the offsets, where $\hat{p}_i$ is the current empirical choice probability and $\hat{\boldsymbol{p}}$ is the empirical choice-probability vector. This is a practical finite-sample adaptation of latent Gaussian categorical modeling [48]; it does not solve the exact multivariate Gaussian choice probabilities.

c) PB3: latent driver with Markov regimes.: PB3 creates a one-dimensional long-memory driver Z_t using either the spectral source above or a Haar-like cascade. The driver is rank-binned into regimes $R_t \in \{1, \dots, M\}$ with target regime-weight vector $\boldsymbol{\pi}$. Conditional on the current regime, symbols follow a first-order Markov chain:

$$\Pr(X_t = a_j \mid X_{t-1} = a_i, R_t = r) = P_{ij}^{(r)}. \tag{12}$$

The local bigram mechanism is therefore explicit through the matrices $P^{(r)}$, while large-scale persistence is inherited from the regime driver. When regimes have nearly identical stationary distributions, one-hot symbol diagnostics may hide the latent LRD; validation therefore reports the latent driver as well as the emitted symbols.

d) PB4: intermittent-map quantization.: PB4 uses a Pomeau-Manneville-style intermittent map

$$Z_{t+1} = (Z_t + Z_t^z) \bmod 1, \qquad z > 1, \tag{13}$$

after a burn-in period. The long laminar episodes near zero create broad finite-scale dependence in the latent driver. `SymbolicLongMemorySequences.jl` then applies the same rank-binning symbolizer as PB1. This method is motivated by intermittent-map approaches to genomic-sequence statistics [49], but the package treats it as a latent finite-sample mechanism rather than an exact closed-form symbolic LRD construction.

TABLE I

IMPLEMENTED SYMBOLIC LRD SYNTHESIS METHODS IN `SymbolicLongMemorySequences.jl`. HERE n IS THE SEQUENCE LENGTH, K IS THE ALPHABET SIZE, d IS A CONFIGURED HISTORY DEPTH, AND I IS THE NUMBER OF PB2 CALIBRATION ITERATIONS. COSTS DESCRIBE GENERATION AFTER CONSTRUCTION FOR THE STANDARD IMPLEMENTATIONS.

ID	Mechanism	Main controls	Typical cost	Main caveat
PB1	Spectral fGn + rank quantization	H, marginal	$O(n \log n)$	local structure induced by quantization
PB2	Latent Gaussian categorical argmax	H, marginal, I	$O(nKI)$	argmax transform is nonlinear and calibrated empirically
PB3	LRD regime driver + Markov chain	H, regime weights, $P^{(r)}$	$O(n \log n + nK)$	one-hot diagnostics may miss hidden regimes
PB4	Intermittent map + rank quantization	map exponent, marginal	$O(n \log n)$	finite laminar scales drive behavior
MB1a	Exact finite-history additive row mixture	β, d, P, marginal	$O(nd)$	finite memory truncates asymptotic LRD
MB1b	Dyadic-bucket additive row mixture	β, d, P, marginal	$O(nK \log d)$	dyadic approximation changes exact history weights
MB1c	Centered additive memory function	β, d, ρ, marginal	$O(nd)$	clipping/renormalization affects strong memory
MB2	Heavy-tailed regime sojourns	α, $P^{(r)}$, Q	$O(nK)$	count-process theory is only a guide for symbols
MB3	Competing fractal renewal streams	tail exponent, rates	$O(nK)$	missing-scale effects can be severe
MB4a	Normalized Hawkes-style excitation	β, d, $\boldsymbol{\mu}$, A	$O(nKd)$	normalized kernel can look short-memory
MB4b	Self-exciting symbol mass	β, d, $\boldsymbol{p}$, b, η, s, A	$O(nKd)$	strong excitation can distort marginals
MB4c	Log-contrast self-exciting mass	β, d, $\boldsymbol{p}$, b, η, s, a, A	$O(nKd)$ or $O(n^2K)$	contrast can cause symbol lock-in
MB5	Power-law lag copy/mutation growth	α, $d_{\max}$, mutation	$O(n \log d)$	growth mechanism can overproduce local patches

The common risk is that nonlinear transformations can change apparent long-memory behavior. The package therefore validates both the latent numerical process and the final one-hot symbolic process.

B. Model-Based Methods

Model-based methods generate symbols directly. They avoid the arbitrary numeric-encoding problem, but they must earn their LRD behavior through the symbolic mechanism itself. This makes their caveats more visible: finite memory, renewal overdispersion, normalization, and copy/mutation patches all show up in validation.

a) MB1a: finite-history LAMP.: MB1a implements a finite-history linear-additive Markov process [38]. Given transition matrix P, target marginal $\boldsymbol{p}$, history depth d, and normalized weights w_ℓ, the next-symbol distribution vector $\boldsymbol{q}_t$ has entries

$$q_t(j) = (1-\epsilon)\sum_{\ell=1}^{m_t} w_{\ell,t} P_{I(X_{t-\ell}),j} + \epsilon p_j + b_{t,j}, \tag{14}$$

where $m_t = \min(d, t-1)$, $I(X)$ maps symbols to indices, and $b_{t,j}$ accounts for missing pre-history mass by assigning it according to $\boldsymbol{p}$. `SymbolicLongMemorySequences.jl` uses power-law weights over observed lags,

$$w_\ell \propto \ell^{-(1+\beta)} \tag{15}$$

for the LAMP-style row mixture. The small innovation parameter ϵ prevents finite-history absorption and preserves a direct marginal-control path. MB1a is exact for the configured finite memory, but it is not a proof that the finite sequence has asymptotic LRD beyond the memory window.

b) MB1b: dyadic-bucket LAMP.: MB1b approximates the MB1a history sum with dyadic lag buckets

$$B_b = \{2^b, \dots, 2^{b+1}-1\}. \tag{16}$$

For each bucket, the generator maintains a count vector $\boldsymbol{h}_b$ of historical symbols and a total bucket weight $W_b = \sum_{\ell \in B_b} w_\ell$. The row-mixture term is approximated by

$$\sum_b W_b \sum_i \hat{h}_{b,i} P_{ij}, \tag{17}$$

where $\hat{h}_{b,i}$ is the empirical symbol mix in bucket b. This preserves the power-law age profile at dyadic resolution while avoiding an explicit scan over all d lags. It is a scalable approximation, not an identical finite-history LAMP.

c) MB1c: centered additive Markov chain.: MB1c follows additive Markov-chain memory-function work [39]–[41]. It writes the next-symbol probability vector as a perturbation around the target marginal:

$$q_t(j) = p_j + \rho \sum_{\ell=1}^{m_t} v_{\ell,t}\left(\mathbf{1}\{X_{t-\ell} = a_j\} - p_j\right), \tag{18}$$

with $v_\ell \propto \ell^{-\beta}$ and strength $\rho \in [0,1]$. The centered form makes the intended recurrence effect clearer than a row mixture: observing a_j in the past increases the future probability of a_j relative to its baseline. Probabilities are clipped and renormalized when finite histories and large ρ would otherwise leave the simplex.

d) MB2: heavy-tailed on/off Markov chain.: MB2 has a regime process R_t with heavy-tailed sojourns. A sojourn length L has Pareto-like tail

$$\Pr(L > x) = (L_{\min}/x)^{\alpha}, \qquad 1 < \alpha < 2, \tag{19}$$

above the scale $L_{\min}$. At regime boundaries, a transition matrix Q selects the next regime. Within a regime,

$$\Pr(X_t = a_j \mid X_{t-1} = a_i, R_t = r) = P_{ij}^{(r)}. \quad (20)$$

Heavy-tailed regime durations create persistent occupancy, while the within-regime matrices control local symbol dynamics. The nominal count-process relationship is $H = (3-\alpha)/2$ for ideal on/off constructions [17], [19].

e) MB3: fractal symbol sequence.: MB3 assigns each symbol a_i an independent event stream with heavy-tailed inter-arrival times. If T_i is the entry of the pending-time vector $\boldsymbol{T}$ for symbol i, the emitted symbol at sequence step t is the one with the smallest pending event time:

$$X_t = a_{\arg\min_i T_i}. \quad (21)$$

After emitting a_i, its pending time is advanced by another heavy-tailed inter-arrival sample. This adapts fractal renewal and shot-noise ideas to a finite alphabet [18], [19]. The known missing-scales problem for naive fractal renewals [22] is a reason to validate the realized scale range rather than rely only on the tail parameter. MB3's rate vector is therefore an asymptotic intensity control: in a finite sequence, heavy-tailed renewal-count fluctuations can dominate the nominal rate ratios.

f) MB4a, MB4b, and MB4c: discrete-time self-exciting symbols.: The first MB4 construction, now denoted MB4a and implemented as `HawkesSymbol`, is a finite-history symbolic analogue of Hawkes word-occurrence models [42]. It follows the Hawkes idea of an immigration or baseline component plus self-excitation from past events. At time t each symbol has an intensity

$$\lambda_{t,j} = \mu_j + \sum_{\ell=1}^{m_t} \bar{g}(\ell) A_{I(X_{t-\ell}),j}, \qquad \bar{g}(\ell) = \frac{(\ell+c)^{-\beta}}{\sum_{r=1}^{d}(r+c)^{-\beta}}, \quad (22)$$

where $m_t = \min(d, t-1)$, $\boldsymbol{\lambda}_t$ is the intensity vector, $\boldsymbol{\mu}$ is a positive baseline vector, and A is a nonnegative excitation matrix. The emitted symbol is sampled from normalized intensities:

$$\Pr(X_t = a_j \mid X_{1:t-1}) = \lambda_{t,j} / \sum_r \lambda_{t,r}. \quad (23)$$

Identity-like A creates bursty repetition; off-diagonal entries create cross-symbol triggering. This is close to a discrete-time marked or multivariate Hawkes intuition, but it differs from a continuous-time Hawkes process because exactly one categorical symbol is emitted at every time step.

Validation exposed a weakness in MB4a. Since $\bar{g}$ has unit mass, the history component has bounded total scale. In the uniform identity-excitation case a first-order linearization has the form

$$Z_t \approx a \sum_{\ell=1}^{d} \bar{g}(\ell) Z_{t-\ell} + \varepsilon_t, \qquad 0 < a < 1, \quad (24)$$

where the baseline and categorical normalization keep a below one. The low-frequency spectrum can then have a plateau rather than a visible power-law pole, so centered one-hot diagnostics can look almost white even when short-run burstiness is present. Increasing the identity-excitation amplitude does not by itself remove this normalization effect.

MB4b, implemented as `SelfExcitingMass`, keeps the discrete-time symbolic setting but changes what is normalized. Let $\boldsymbol{p}$ be the target/default marginal, let $b \geq 0$ be a default mass, let $\eta \geq 0$ be an excitation strength, and let $s \geq 0$ be a smoothing floor. MB4b forms unnormalized masses

$$M_{t,j} = b p_j + \eta \left(s p_j + \sum_{\ell=1}^{m_t} (\ell+c)^{-\beta} A_{I(X_{t-\ell}),j} \right), \quad (25)$$

and only then samples

$$\Pr(X_t = a_j \mid X_{1:t-1}) = M_{t,j} / \sum_r M_{t,r}. \quad (26)$$

The key difference is that the power-law kernel is not normalized in the memory mass. For $0 < \beta < 1$, the represented memory mass grows like $d^{1-\beta}$ over the finite range, allowing self-excitation to dominate the small default component once enough history exists. The default and smoothing terms are retained to make the startup distribution normalizable and to prevent symbols from becoming permanently unreachable. MB4b is therefore best viewed as our discrete-time categorical adaptation of Hawkes-style self-excitation, rather than a standard Hawkes model already present in the literature. The closest literature precedent is MB4a's Hawkes/word-recurrence motivation; MB4b is the package modification introduced because the traditional normalized categorical translation gave weak LRD diagnostics.

The remaining weakness of MB4b is that the final categorical normalization still uses the total mass $\sum_r M_{t,r}$. When the identity-excitation model has nearly uniform symbol usage, much of the accumulated long-memory mass is common mode across symbols. MB4c, implemented as `LogitSelfExcitingMass`, keeps the same unnormalized mass $M_{t,j}$ but converts it to a centered log contrast before sampling. Let $a \geq 0$ be a contrast strength and $\epsilon > 0$ a small logarithmic floor. Define

$$\theta_{t,j} = a \left[\log(M_{t,j} + \epsilon) - \sum_r p_r \log(M_{t,r} + \epsilon) \right]. \quad (27)$$

MB4c then samples according to

$$\Pr(X_t = a_j \mid X_{1:t-1}) = \frac{p_j \exp(\theta_{t,j})}{\sum_r p_r \exp(\theta_{t,r})}. \quad (28)$$

The subtraction removes common-mode growth in the history mass before the final probability normalization. Conceptually, the power-law kernel need not have a finite memory limit; the implementation parameter d is a computational truncation, and `d = nothing` in MB4c uses all previous generated symbols. The offset c is likewise not a second LRD parameter: it regularizes short lags through $(\ell+c)^{-\beta}$, while β controls the asymptotic power-law decay. Preliminary validation shows the expected tradeoff. Increasing a can steepen the low-frequency spectrum, but too large a contrast creates symbol lock-in and poor finite-sample marginals. We therefore treat MB4c as a new experimental variant and use it to expose the remaining design problem rather than as a final closed-form solution.

g) MB5: duplication-mutation growth.: MB5 starts from an iid seed sequence and grows by copying a previous symbol at a random lag. The lag D_t is drawn from a truncated power law,

$$\Pr(D_t = d) \propto d^{-\alpha}, \qquad 1 \le d \le d_{\max}. \tag{29}$$

With probability $1-\mu$, the next symbol is copied, $X_t = X_{t-D_t}$; with probability μ, it is mutated by drawing from the target marginal or an alternative symbol distribution. This is motivated by duplication-mutation and expansion-modification ideas in DNA-like symbolic sequences [26], [43], [44]. `SymbolicLongMemorySequences.jl` uses the power-law lag-copy version because validation showed that a block-length-only version produced local patches without the intended symbolic autocorrelation decay.

V. Validation Results

A. ACF and Spectra for Symbolic Sequences

The introduction argues that symbolic LRD claims depend on the diagnostic used. The first validation question is therefore how to compute an ACF for objects that are not numbers. A symbolic sequence has no canonical numerical scale. For example, replacing $\{\mathtt{A}, \mathtt{C}, \mathtt{G}, \mathtt{T}\}$ by $\{1, 2, 3, 4\}$ would make the distance between A and T three times the distance between A and C, even though no such distance is part of the alphabet. `SymbolicLongMemorySequences.jl` therefore validates symbolic dependence through centered indicator processes, following the spirit of Voss's DNA indicator method [23].

Let $\boldsymbol{X}_{1:n} = (X_1, \ldots, X_n) \in \mathcal{A}^n$, where $\mathcal{A} = \{a_1, \ldots, a_K\}$. For each symbol a_j, define the binary indicator and its centered version

$$B_t(a_j) = \mathbf{1}\{X_t = a_j\}, \qquad Y_t(a_j) = B_t(a_j) - \bar{B}(a_j), \tag{30}$$

where $\bar{B}(a_j) = n^{-1} \sum_{t=1}^{n} B_t(a_j)$. The sample autocovariance for symbol a_j at lag h is then

$$\hat{\gamma}_j(h) = \frac{1}{n-h} \sum_{t=1}^{n-h} Y_t(a_j) Y_{t+h}(a_j), \qquad h = 0, \ldots, n-1, \tag{31}$$

and the corresponding autocorrelation is

$$\hat{\rho}_j(h) = \hat{\gamma}_j(h) / \hat{\gamma}_j(0), \tag{32}$$

when $\hat{\gamma}_j(0) > 0$. Symbols with zero empirical variance are omitted. The reported symbolic ACF is an average across the remaining symbol indicators, usually after retaining positive values for log-log display:

$$\hat{\rho}_{\mathrm{sym}}(h) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \hat{\rho}_j(h), \qquad \mathcal{J} = \{j : \hat{\gamma}_j(0) > 0\}. \tag{33}$$

This is not the only possible symbolic diagnostic, but it has a clear interpretation: it asks whether occurrences of the same symbol remain correlated over long lags.

The spectral diagnostic uses the same centered indicator series. For Fourier frequencies $f_m = m/n$, a simple periodogram for symbol a_j is

$$\hat{S}_j(f_m) = \frac{1}{n} \left| \sum_{t=1}^{n} Y_t(a_j) \exp(-2\pi i f_m t) \right|^2. \tag{34}$$

`SymbolicLongMemorySequences.jl` averages these symbol-wise periodograms across $j \in \mathcal{J}$, then plots the result on log-log axes. A symbolic sequence with covariance-like LRD in its indicator processes should show slowly decaying positive ACF values and elevated low-frequency spectral power, subject to finite-sample noise.

As a toy example, take

$$\boldsymbol{X}_{1:8} = (a, b, a, a, c, b, a, c).$$

For symbol a,

$$\boldsymbol{B}(a) = (1, 0, 1, 1, 0, 0, 1, 0), \qquad \bar{B}(a) = 4/8,$$

so

$$\boldsymbol{Y}(a) = (1/2, -1/2, 1/2, 1/2, -1/2, -1/2, 1/2, -1/2).$$

The lag-one autocovariance contribution is the average of

$$\boldsymbol{c}_1 = (-1/4, -1/4, 1/4, -1/4, 1/4, -1/4, -1/4),$$

which is negative for this short sequence; the symbol a tends not to repeat at lag one except for the pair at positions 3 and 4. The same calculation is performed for b and c, and the symbol-wise ACFs are averaged. For a real validation run this procedure is applied to much longer sequences and many lags; the toy example is only meant to show the transformation from symbols to numerical diagnostics.

The plotted dashed reference lines show nominal power-law behavior and finite-sample interpretation limits. They are visual guides, not estimator guarantees. Centered one-hot diagnostics also have limitations: they can miss dependence that appears only in higher-order patterns, changing transition matrices, or recurrence-time distributions. This limitation reflects the problem rather than the implementation: symbolic LRD cannot be reduced to one statistic, so the diagnostic must be stated beside the generated data.

B. Marginal-Control Validation

Because the paper treats synthetic data as a controlled experimental object, LRD diagnostics are only meaningful if the generator also respects the requested symbol frequencies well enough for the intended experiment. `SymbolicLongMemorySequences.jl` therefore includes a separate marginal-control validation. The paper-facing case uses a uniform categorical target with $K = 8$, generates $n = 100000$ symbols for each replicate, and drops the first and last 10% before computing frequencies. The trimmed window reduces sensitivity to initialization and terminal edge effects, leaving 80000 observations per replicate in the reported run.

The uniform target has two advantages. First, if a generator can produce a fine uniform categorical distribution, other marginals can be approximated by aggregating fine bins; direct nonuniform targets are still tested elsewhere in the validation script, but the uniform case gives the most comparable shared diagnostic. Second, the uniform case avoids small expected cell counts. Classical chi-squared frequency tests become unreliable when bins have low expected counts, a common problem for skewed symbolic distributions.

For each replicate, the validation computes

$$\chi^2_{\text{obs}} = \sum_{j=1}^{K} \frac{(N_j - m/K)^2}{m/K}, \tag{35}$$

where m is the trimmed sample size and N_j is the count for symbol a_j. The reported p-values use the iid multinomial chi-squared reference with $K-1$ degrees of freedom. This reference is conservative as an interpretation aid: the generated sequences are dependent, so the p-values are frequency diagnostics rather than exact hypothesis-test probabilities. The issue is especially important for LRD generators because empirical averages can converge more slowly than the iid multinomial model assumes. Thus a strict chi-squared test may reject too often even when the generator has the intended long-run marginal, although large values remain indicative of a control problem.

The validation table also reports an approximate effective-sample-size (ESS) correction. For symbol a_j, define the centered indicator $Y_{j,t} = \mathbb{1}\{X_t = a_j\} - \hat{p}_j$ on the trimmed sequence. Let $\hat{\rho}_j(h)$ be its sample autocorrelation at lag h. We use an initial-positive truncation. That is, $h_j^\star$ is the largest lag before the first non-positive value in the sequence $\hat{\rho}_j(1), \hat{\rho}_j(2), \ldots$, and

$$\hat{\tau}_j = 1 + 2\sum_{h=1}^{h_j^\star} \hat{\rho}_j(h), \tag{36}$$

with $\hat{\tau}_j \geq 1$, and define

$$\hat{m}_{\text{eff}} = \min_{1\leq j\leq K} \frac{m}{\hat{\tau}_j}. \tag{37}$$

The ESS-adjusted statistic is then

$$\chi^2_{\text{ESS}} = \chi^2_{\text{obs}} \frac{\hat{m}_{\text{eff}}}{m}. \tag{38}$$

This scalar correction is not a full dependent multinomial theory. It uses the smallest symbol-level ESS, so it is conservative, but it helps identify cases where an apparently large frequency deviation is partly explained by long dependence in the symbol indicators. For MB1a and MB1b, the marginal-control validation uses the package's repeat-biased identity/dyad transition helper, with repeat probability 0.4. This keeps a history-repetition mechanism while making $\boldsymbol{p}$ the stationary distribution of the transition matrix. A pure identity matrix is still a useful stress case, but it can preserve early finite-sample imbalances for a long time. The retained table also reports full-sequence marginal error, because some methods calibrate or rank-bin the entire generated sample while the trimmed interior window is allowed to fluctuate.

A better formal test would calibrate the null distribution under dependence. Possible extensions include block or subsampling tests, or a parametric Monte Carlo envelope generated from the same configured generator. These extensions are left for later work. The simple chi-squared reference and the ESS-adjusted diagnostic remain useful because they are transparent, comparable across methods, and sensitive to obvious marginal-control failures.

The table shows three different phenomena. PB1 and PB4 use rank binning and therefore match the full sample exactly in this uniform case, but the interior trimmed window can wander because the sequence is dependent. PB2 similarly calibrates the whole generated sample very closely, while the trimmed window has larger deviations. Methods with explicit iid or stationary uniform mechanisms, including PB3, MB1a, MB1b, MB1c, MB2, MB4a, MB4b, MB4c, and MB5 in the tested configurations, mostly match the marginal closely in total-variation terms. MB3 is the clear exception. Its independent heavy-tailed renewal streams show the largest replicate variability in both the full and trimmed samples, and the ESS correction does not change the conclusion: the estimated one-hot ESS is the full trimmed length in this run. This indicates that the failure is not primarily a short-lag serial-correlation correction problem. Rather, it is a renewal-count overdispersion problem: over a finite horizon, one symbol stream can realize many short gaps while another realizes one or more very long gaps, even when the nominal rates are equal. The resulting marginal imbalance can be much larger than iid multinomial theory expects. These results are not LRD estimates; they are control diagnostics that identify when a generator's long-memory mechanism interferes with a simple marginal target.

C. Toward One-Step Markov Validation

Marginals are only the first local control. The same validation philosophy extends to one-step Markov behavior, but direct matrix testing is more data-hungry. A $K \times K$ transition matrix has K^2 cells, and the effective count in row i is proportional to the number of visits to symbol a_i. For large or skewed alphabets, many rows can have too little data for a useful cell-wise chi-squared comparison.

A practical validation shortcut is to test interpretable low-dimensional contrasts of the transition matrix before attempting a full matrix test. Useful summaries include the repeat probability $\Pr(X_t = X_{t-1})$, stationary-weighted row total variation, selected rows for scientifically important symbols, grouped transitions after aggregating symbols into coarser classes, and projections $u^\top(P-\hat{P})v$ for a small set of contrast vectors u and v. These diagnostics do not replace full row-by-row validation for small K or very long runs, but they provide a shorter path for detecting whether local persistence, cycling, or group-level movement has the intended strength.

The validation plots are mainly comparative. PB1 is the simplest baseline: the latent spectral construction is visible and rank quantization often leaves a recognizable symbolic power-law range. PB2 generally preserves a useful long-memory signal, but the argmax transform and marginal-offset calibration make the symbolic diagnostics less transparent than PB1. PB3 separates the latent driver from the emitted Markov chain; this is useful because the latent diagnostics can remain convincing even when the symbol indicators are partly blind to regime persistence. PB4 illustrates a different phenomenology: dependence is created by intermittent laminar episodes rather than by a Gaussian spectral pole.

Among model-based methods, MB1b and MB1c are intended as the most direct symbolic additive-memory baselines, with MB1a kept as the exact finite-history reference. MB2

TABLE II

UNIFORM MARGINAL VALIDATION FOR $K = 8$, $n = 100000$, 20 REPLICATES, AND A TRIMMED WINDOW OF 80000 OBSERVATIONS PER REPLICATE. THE IID P-VALUES USE THE MULTINOMIAL CHI-SQUARED REFERENCE. THE ESS P-VALUES SCALE THE STATISTIC BY THE CONSERVATIVE EFFECTIVE SAMPLE SIZE ESTIMATED FROM ONE-HOT SYMBOL AUTOCORRELATIONS. BOTH SHOULD BE READ AS DIAGNOSTICS UNDER DEPENDENCE. TV DENOTES TOTAL VARIATION DISTANCE FROM THE TARGET MARGINAL.

Method	Mean χ^2_{obs}	Median iid p	Median ESS p	Full TV	Trim TV
PB1 `SpectralFGN`	39.65	0.040	0.997	0.0000	0.0074
PB2 `LGCM`	87.69	4.55×10^{-14}	0.966	0.0001	0.0135
PB3 `WaveletMarkov`	6.84	0.514	0.525	0.0034	0.0036
PB4 `IntermittentMapSymbols`	25.18	0.065	1.000	0.0000	0.0053
MB1a `LAMP`	18.92	0.021	0.445	0.0053	0.0060
MB1b `DyadicLAMP`	15.52	0.035	0.486	0.0048	0.0055
MB1c `CalibratedAdditiveMarkov`	7.43	0.493	0.508	0.0035	0.0040
MB2 `OnOffMarkov`	6.33	0.567	0.583	0.0032	0.0035
MB3 `FSS`	203.01	5.48×10^{-17}	5.48×10^{-17}	0.0160	0.0175
MB4a `HawkesSymbol`	7.43	0.493	0.508	0.0035	0.0040
MB4b `SelfExcitingMass`	7.43	0.493	0.508	0.0035	0.0040
MB4c `LogitSelfExcitingMass`	7.43	0.493	0.508	0.0035	0.0040
MB5 `DuplicationMutation`	6.81	0.531	0.545	0.0033	0.0035

and MB3 are mechanism-oriented renewal/regime baselines and are particularly useful when recurrence or burst durations are the quantity of interest. The weaker MB4a and MB5 results are included because MB4a's normalized finite-history excitation can have a nearly white symbol spectrum, and MB5's copy/mutation growth can concentrate dependence into local duplication structure rather than a clean one-hot power law. MB4b is the replacement candidate motivated by this failure mode, but its finite-sample behavior still requires validation. These cases show why symbolic LRD generators require empirical diagnostics in addition to parameter labels.

VI. BENCHMARK RESULTS AND DISCUSSION

The benchmark section supports the same operational narrative: what does each mechanism cost when it is used as a generator rather than as a mathematical idea? The measurements are generated with `BenchmarkTools.jl` in a separate environment. The extended scaling mode uses sequence lengths from 10^2 to 10^6 and averages each timing over multiple independently seeded syntheses. The resulting histograms and scaling plots are written under `benchmark/results/` and summarized in `benchmark/RESULTS.md`.

The expected complexity classes are visible, although implementation constants are also important. Spectral methods pay for FFTs and sorting and scale near $O(n \log n)$. Finite-history additive and self-exciting methods pay for their memory window. Dyadic buckets trade exact history access for a scalable approximation, but the current MB1b implementation still has a large constant factor. The table should therefore be read as engineering evidence, not as an asymptotic result.

The benchmark also reinforces the validation story. MB2 and MB3 are fast because they avoid long history scans, but their interpretation is tied to renewal and regime mechanisms. PB1 is a strong baseline because the latent second-order target is simple, even though it pays for FFTs and allocation. PB2 becomes expensive for large alphabets because it generates one latent stream per symbol and calibrates offsets. Speed only matters when the generator expresses the intended symbolic dependence, and a convincing finite-sample plot is not a proof of asymptotic LRD. Both pieces of evidence are needed.

VII. LIMITATIONS AND FUTURE WORK

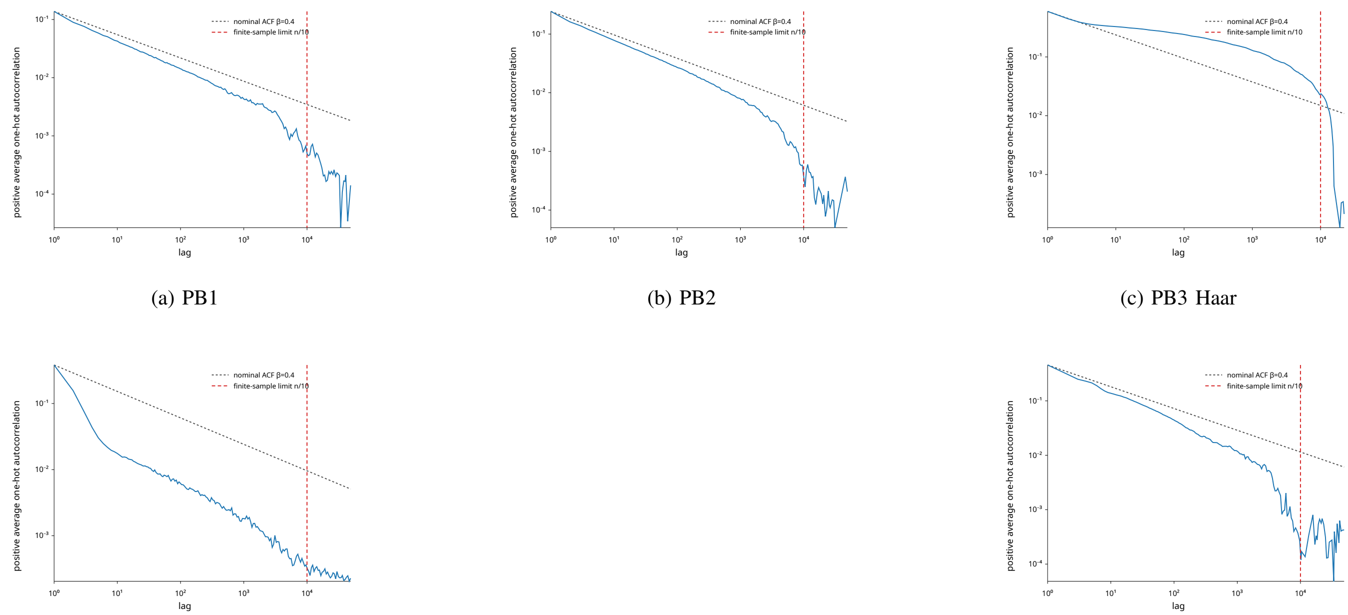
The main limitation is also the main reason for the package design: symbolic LRD has several legitimate meanings. Centered one-hot ACFs and spectra are transparent and close to the DNA-indicator literature, but they can understate regime-based, recurrence-based, or higher-order dependence. Future validation should add recurrence-time, count-variance, block-entropy, and local-structure diagnostics without turning `SymbolicLongMemorySequences.jl` itself into an estimator package.

Several generators are intentionally research prototypes. MB4a is retained as a cautionary normalized-excitation baseline rather than a recommended generator. MB4b and MB4c are stronger candidates, but still need a cleaner route from power-law excitation to a stable symbolic spectral slope, perhaps through a branching-process or count-process formulation before final categorical sampling. MB5 needs a better separation between local duplication patches and long-lag copying. MB3 needs a marginal-stabilized renewal construction. A promising variant would retain heavy-tailed renewal or intensity dynamics but add finite-sample marginal calibration, for example by generating candidate renewal events and then thinning, accepting, or allocating emissions against target symbol counts. That should probably be exposed as a calibrated MB3 variant rather than silently replacing the current FSS model, because the current model's scientific identity is the competition among independent fractal renewal streams. PB3 would benefit from calibration that maximizes contrast among regime stationary distributions while preserving requested local transition matrices.

This is why the package is designed to expose weaknesses rather than hide them. A symbolic LRD method should fail visibly when its mechanism does not support the claim being made. The next iteration should make those failures more diagnostic and, where possible, turn them into better generators.

VIII. REPRODUCIBILITY ARTIFACTS

The paper is built from retained scripts and result artifacts. Validation figures are produced by scripts under



(a) PB1 (b) PB2 (c) PB3 Haar

(d) PB3 spectral (e) PB4

Fig. 1. Production ACF diagnostics for property-based methods. Each subplot is the centered one-hot symbolic ACF with finite-length and target power-law guides; latent ACFs and spectral-density panels are generated by the validation scripts and synced into the paper figure tree, but are omitted here to save space.

TABLE III
BENCHMARKTOOLS.JL GENERATION TIMINGS FOR $K = 8$ AND $n = 10^6$ ON THE DEVELOPMENT MACHINE. VALUES ARE PER-SYNTHESIS AVERAGES WITHIN EACH BENCHMARKTOOLS.JL TRIAL; THE BENCHMARK SCRIPT USED TEN SYNTHESES PER TRIAL. THESE NUMBERS ARE MACHINE-SPECIFIC AND SHOULD BE INTERPRETED PRIMARILY AS RELATIVE COSTS FOR THE CONFIGURED STANDARD CASES.

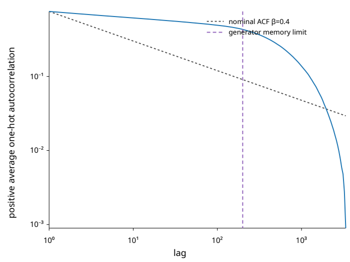
Method	Time (ms)	Relative	Trial allocations	Trial memory (MiB)
MB2 OnOffMarkov	13.301	1.00	480	76.33
MB5 DuplicationMutation	22.293	1.68	90	153.35
MB3 FSS	23.347	1.76	60	76.30
PB4 IntermittentMapSymbols	102.039	7.67	320	267.04
PB1 SpectralFGN	117.242	8.81	490	648.52
PB3 WaveletMarkov spectral	150.058	11.28	730	724.83
MB1c CalibratedAdditiveMarkov	270.006	20.30	80	152.59
MB4a HawkesSymbol	280.897	21.12	80	152.59
MB1a LAMP	283.760	21.33	80	152.59
MB4b SelfExcitingMass	284.050	21.36	80	152.59
PB2 LGCM	439.012	33.01	1630	4348.82
MB4c LogitSelfExcitingMass	1390.853	104.57	100	152.59
MB1b DyadicLAMP	1557.556	117.10	130	762.98

validation/, with aggregate tables and plots written under validation/results/lrd_diagnostics/. Benchmark tables and plots are produced by benchmark/benchmarks.jl in its separate BenchmarkTools.jl project, with extended scaling controlled by explicit environment flags so rare runs are not part of ordinary tests. The paper and annotated bibliography are built by Paper_s4/Makefile, which writes LaTeX intermediates to Paper_s4/Tex/. The makefile also rsyncs the production validation and benchmark plots from the package repository into Paper_s4/Plots/ before compiling.

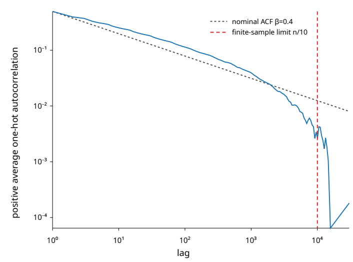
Reference provenance is part of the artifact. Downloaded papers are kept outside the public repository, while Paper_s4/DOWNLOAD.mn records match status and private-store paths. The BibTeX database was checked with PaperFetch.jl in online check-only mode; the check was used to correct clear metadata conflicts without downloading additional PDFs. This keeps the public repository lightweight while preserving an audit trail for the literature used to construct and interpret the generators.

IX. CONCLUSION

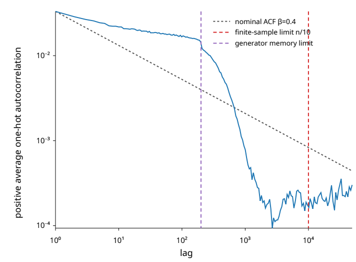
SymbolicLongMemorySequences.jl is a reproducible workbench for synthesizing long-memory symbolic data without treating symbols as numbers. The package provides a common API, explicit mechanisms, validation scripts, and benchmarks for property-based and model-based generators. It is useful for LRD estimator testing, entropy and string-method test sets, controlled machine-learning stress tests,



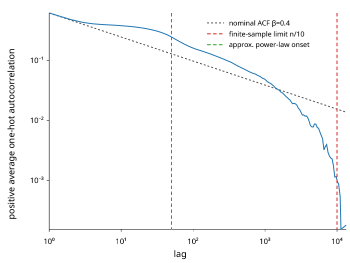
(a) MB1a



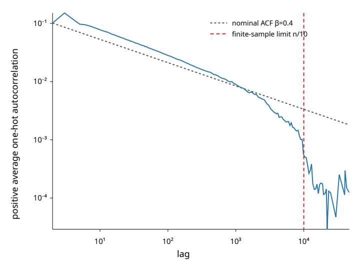
(b) MB1b



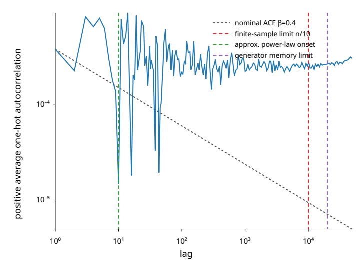
(c) MB1c



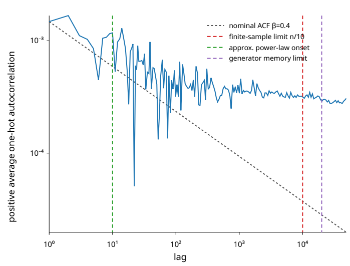
(d) MB2



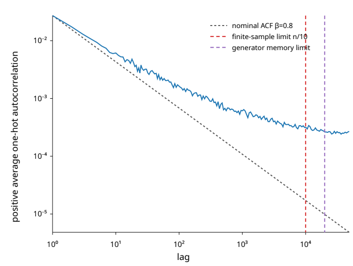
(e) MB3



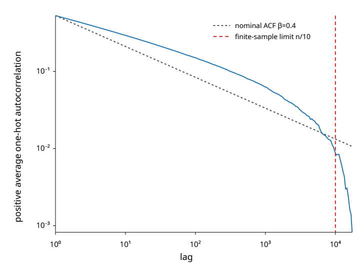
(f) MB4a



(g) MB4b



(h) MB4c



(i) MB5

Fig. 2. Production ACF diagnostics for model-based methods. The panels use the same title-free production plot style as Fig. 1, with method names supplied by subcaptions rather than embedded titles.

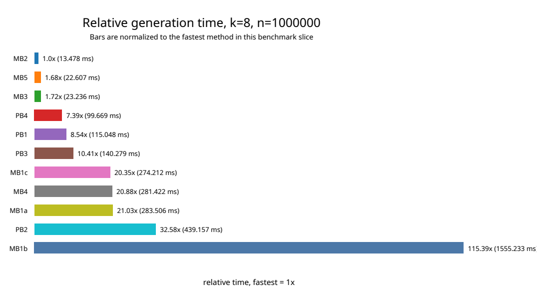


Fig. 3. Relative generation time for alphabet size $K = 8$ and sequence length $n = 10^6$. Times are normalized to the fastest method in this benchmark run.

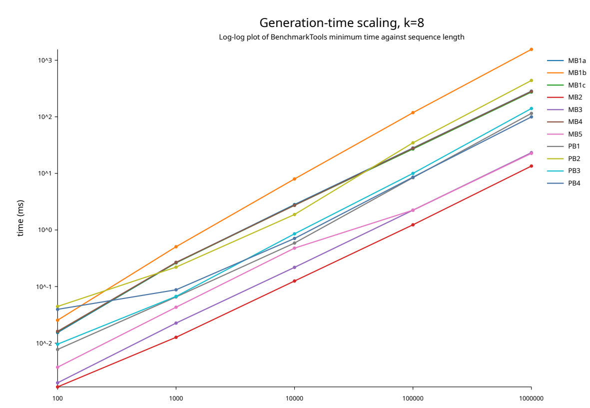


Fig. 4. Generation-time scaling with sequence length for alphabet size $K = 8$ on log-log axes.

synthetic event-log generation, privacy-preserving simulation, anomaly and fake-source detection studies, and workload modeling for symbolic streams.

The methodological point, and the narrative thread of the paper, is that for symbolic LRD a parameter label is not enough. We need to know the marginal, the local structure, the mechanism, the transformation, the finite memory limit, and the diagnostic used to judge the output. Those pieces need to be explicit; no single generator supplies all of them. Future work should strengthen correlation-calibrated symbolic generators, add exact numerical latent sources, and connect the validation evidence to external estimator packages while keeping SymbolicLongMemorySequences.jl focused on reproducible synthesis.

Appendix A
Slow Convergence of Averages Under LRD

Long-range dependence changes how empirical averages stabilize. For an iid or short-range dependent scalar process, the variance of a sample mean typically scales as $O(n^{-1})$, giving the familiar $n^{-1/2}$ standard-error rate. If a stationary scalar process Z_t has autocovariance

$$\gamma(h) \sim c_\gamma h^{-\beta}, \qquad 0 < \beta < 1, \tag{39}$$

then the covariance terms in the sample mean do not sum to a constant. Instead,

$$\operatorname{Var}(\bar{Z}_n) = \frac{1}{n^2}\left[n\gamma(0) + 2\sum_{h=1}^{n-1}(n-h)\gamma(h)\right] \tag{40}$$

$$\sim Cn^{-\beta}, \tag{41}$$

for a positive constant C when the tail coefficient is positive [4]–[6]. Thus the standard error decays as $n^{-\beta/2}$, rather than $n^{-1/2}$. In Hurst-parameter notation, $\beta = 2 - 2H$, so the standard error behaves like n^{H-1}. For $H = 0.8$, for example, the standard error is of order $n^{-0.2}$, so increasing the sequence length by a factor of 100 only reduces the typical average error by about a factor of $100^{-0.2} \approx 0.40$, not by a factor of 10.

The same calculation applies directly to marginal estimation for symbolic sequences. For alphabet symbol a_j, define the indicator

$$I_{j,t} = \mathbf{1}\{X_t = a_j\}, \qquad \hat{p}_j = \frac{1}{n}\sum_{t=1}^{n} I_{j,t}. \tag{42}$$

If the centered indicator process $I_{j,t} - p_j$ has autocovariance $\gamma_j(h) \sim c_j h^{-\beta_j}$, then

$$\operatorname{Var}(\hat{p}_j) \sim C_j n^{-\beta_j}. \tag{43}$$

This is much larger than the iid multinomial approximation $p_j(1-p_j)/n$ at large n. Equivalently, the effective number of independent observations grows sublinearly; up to constants, an LRD sequence with exponent β can behave like only $n_{\text{eff}} \propto n^{\beta}$ independent observations for mean estimation. The exact constant depends on the full autocorrelation function, not only on its exponent, so this scaling should be read as an order-of-magnitude guide rather than a replacement for a generator-specific uncertainty calculation.

This has two practical consequences for the marginal-control validation in Section V. First, a histogram from a long symbolic sequence can still show visible departures from the requested marginal even when the generator has the right long-run target. These departures need not disappear at the iid $n^{-1/2}$ rate. Second, a classical multinomial chi-squared test treats the counts as though they came from n independent draws. For LRD sequences this can make p-values too small and can overstate the evidence against the target marginal. The ESS-adjusted statistic used here is an approximate correction: it reduces the nominal sample size using the observed one-hot autocorrelation. It remains approximate, because a full dependent multinomial test would require the joint covariance structure of the count vector, but it is a guard against interpreting slow LRD convergence as an immediate marginal-control failure.

Uniform targets are still valuable in this setting. They keep expected bin counts away from zero and make deviations visually comparable across symbols. However, the right interpretation is diagnostic rather than absolute: rank-binned methods may match the full-sample marginal almost exactly while the trimmed interior fluctuates, and process-based methods may require many independent replicates before their average histogram settles. Trimming the first and last 10% reduces startup and terminal edge effects, but it does not remove the slow convergence caused by long memory itself.

A. Uniform Marginal Histograms

The validation script writes a complete histogram grid for all methods, but the paper includes only the worst trimmed-total-variation case from the current validation run. The remaining methods are summarized by the marginal table in the main validation section and have smaller deviations from the target uniform marginal. Figure 5 shows the mean trimmed empirical symbol frequencies for that worst case; error bars show one replicate standard deviation and the dashed red line marks the intended frequency $1/K$.

Appendix B
Transformations of LRD Processes

Let Z_t be a real-valued, zero-mean stationary process with autocovariance

$$\gamma_Z(k) \sim ck^{-\beta}, \qquad 0 < \beta < 1. \tag{44}$$

For a linear transformation $Y_t = aZ_t + b$ with $a \neq 0$,

$$\gamma_Y(k) = a^2\gamma_Z(k), \tag{45}$$

so the LRD exponent is unchanged.

A. Linear Maps and Vector Encodings

For vector-valued encodings, let $\boldsymbol{Y}_t = B\boldsymbol{e}(X_t)$, where $\boldsymbol{e}(a_i)$ is the ith coordinate vector and B is a fixed matrix. If the centered indicator covariance matrix

$$\Gamma_I(k) = \mathbb{E}\left[(\boldsymbol{e}(X_t) - \boldsymbol{p})(\boldsymbol{e}(X_{t+k}) - \boldsymbol{p})^\top\right] \tag{46}$$

has entries with a common power-law tail, then

$$\Gamma_{\boldsymbol{Y}}(k) = B\Gamma_I(k)B^\top. \tag{47}$$

A scalar numerical encoding is the special case $B = \boldsymbol{b}^\top$. This can preserve a tail if $\boldsymbol{b}$ has nonzero projection on the long-memory eigenspace of $\Gamma_I(k)$, but it can also cancel or hide dependence. This is one reason arbitrary scalar DNA walks or word-index encodings can be misleading, and why SymbolicLongMemorySequences.jl defaults to centered one-hot diagnostics [23].

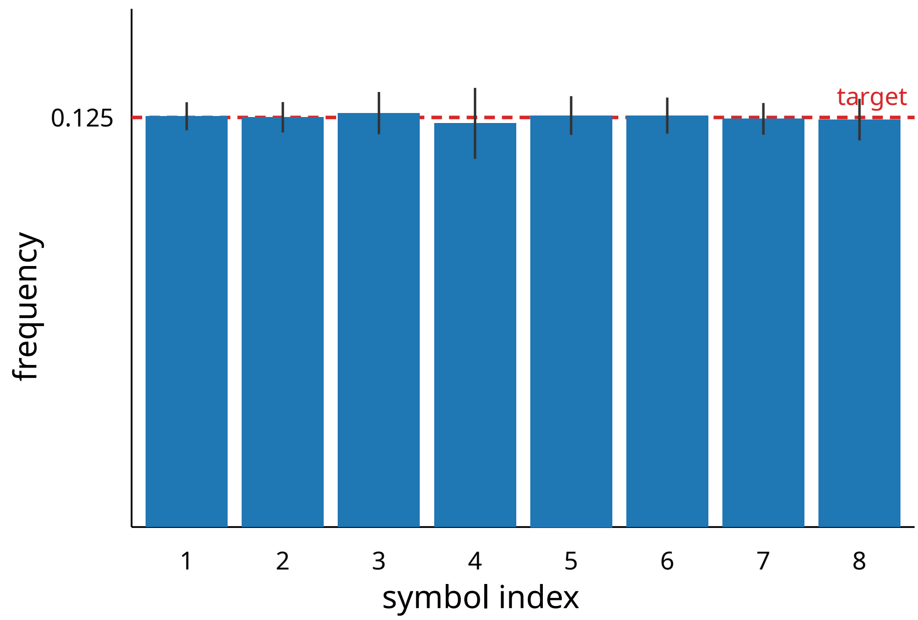


Fig. 5. Worst-case uniform marginal validation histogram for $K = 8$ in the current validation run. The full generated grid is retained as a validation artifact, but only this panel is included here because the other methods are closer to the target marginal.

B. Gaussian Nonlinear Transformations

For nonlinear transformations, preservation depends on both the function and the law of the latent process. The cleanest classical result is for a stationary standard Gaussian process Z_t with correlation

$$r(k) \sim ck^{-\beta}. \tag{48}$$

Let h be square-integrable under the standard normal law and centered so that $\mathbb{E}h(Z) = 0$. Its Hermite expansion is

$$h(x) = \sum_{q=0}^{\infty} \frac{J_q}{q!} H_q(x), \qquad J_q = \mathbb{E}[h(Z)H_q(Z)], \tag{49}$$

where H_q is the qth Hermite polynomial. The Hermite rank m is the first positive q with $J_q \neq 0$. Using orthogonality of Hermite polynomials for joint Gaussian variables,

$$\mathrm{Cov}(h(Z_t), h(Z_{t+k})) = \sum_{q=m}^{\infty} \frac{J_q^2}{q!} r(k)^q. \tag{50}$$

Thus, when the first nonzero term dominates,

$$\mathrm{Cov}(h(Z_t), h(Z_{t+k})) \sim \frac{J_m^2}{m!} c^m k^{-m\beta}. \tag{51}$$

The transformed process remains covariance-LRD only when $m\beta < 1$. Otherwise the covariance tail becomes summable even though the latent process was LRD. This Hermite-rank principle underlies classical non-central limit theory for nonlinear functions of Gaussian LRD processes [50]–[53].

C. Thresholds, Quantization, and Indicators

Quantization is a nonlinear map. For a binary threshold

$$Y_t = \mathbf{1}\{Z_t \leq q\} - \Phi(q), \tag{52}$$

the first Hermite coefficient is

$$J_1 = \mathbb{E}[Y_t Z_t] = -\phi(q), \tag{53}$$

which is nonzero for finite q. Hence a single Gaussian threshold usually has Hermite rank one and preserves the covariance exponent:

$$\mathrm{Cov}(Y_t, Y_{t+k}) \sim \phi(q)^2 r(k). \tag{54}$$

For a multi-bin quantizer, the centered indicator for bin $(q_{i-1}, q_i]$ has first Hermite coefficient

$$J_{1,i} = \mathbb{E}\left[(\mathbf{1}\{q_{i-1} < Z \leq q_i\} - p_i) Z\right] = \phi(q_{i-1}) - \phi(q_i), \tag{55}$$

with the convention that $\phi(\pm\infty) = 0$. Interior symmetric bins can have small or zero first coefficient, in which case higher Hermite terms can dominate at observable scales. Rare bins can also have weak finite-sample diagnostics because their indicator variance is small and long-lag estimates have high relative noise.

This analysis explains why PB1 can often preserve a visible power-law in common symbol indicators, while some symbols or transformations may look weaker. It also explains why PB2's argmax transform is harder to reason about: the emitted indicator is a nonlinear function of a K-dimensional Gaussian vector, and its effective Hermite rank depends on the offsets, marginal probabilities, and symmetries among latent streams.

D. Markov Regime Transformations

PB3 introduces another layer: a latent LRD driver determines a regime process R_t, and symbols are emitted through regime-specific Markov chains. If R_t has long memory and the stationary distribution vectors $\boldsymbol{\pi}^{(r)}$ of the matrices $P^{(r)}$

differ across regimes, then a symbol indicator can inherit part of the regime dependence through

$$\mathbb{E}[\mathbf{1}\{X_t = a_j\} \mid R_t = r] \approx \pi_j^{(r)}. \tag{56}$$

However, if all regimes have the same stationary distribution, the first-order symbol indicators may be nearly blind to the regime process even though higher-order local structure is changing. In that case, diagnostics based only on one-hot autocorrelation can understate the dependence created by the latent driver. This motivates validating both latent regimes and emitted symbols.

E. Non-Gaussian Latent Processes

The Hermite argument does not apply directly to intermittent maps or heavy-tailed renewal mechanisms. For such processes, transformations can alter dependence through changes in laminar-state visibility, tail truncation, and finite-sample scale range. Intermittent-map quantization may emphasize laminar episodes if a bin isolates the sticky region near zero, or weaken them if rank binning spreads laminar values across symbols. Heavy-tailed on/off and renewal models can preserve count-process LRD while individual symbol autocorrelations are noisy or dominated by rare-event effects.

The practical conclusion for `SymbolicLongMemorySequences.jl` is conservative. The package does not claim that a nominal parameter always determines the finite symbolic ACF exponent. Instead it records the mechanism, exposes latent outputs where they exist, and produces validation evidence at the numerical and symbolic levels.

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